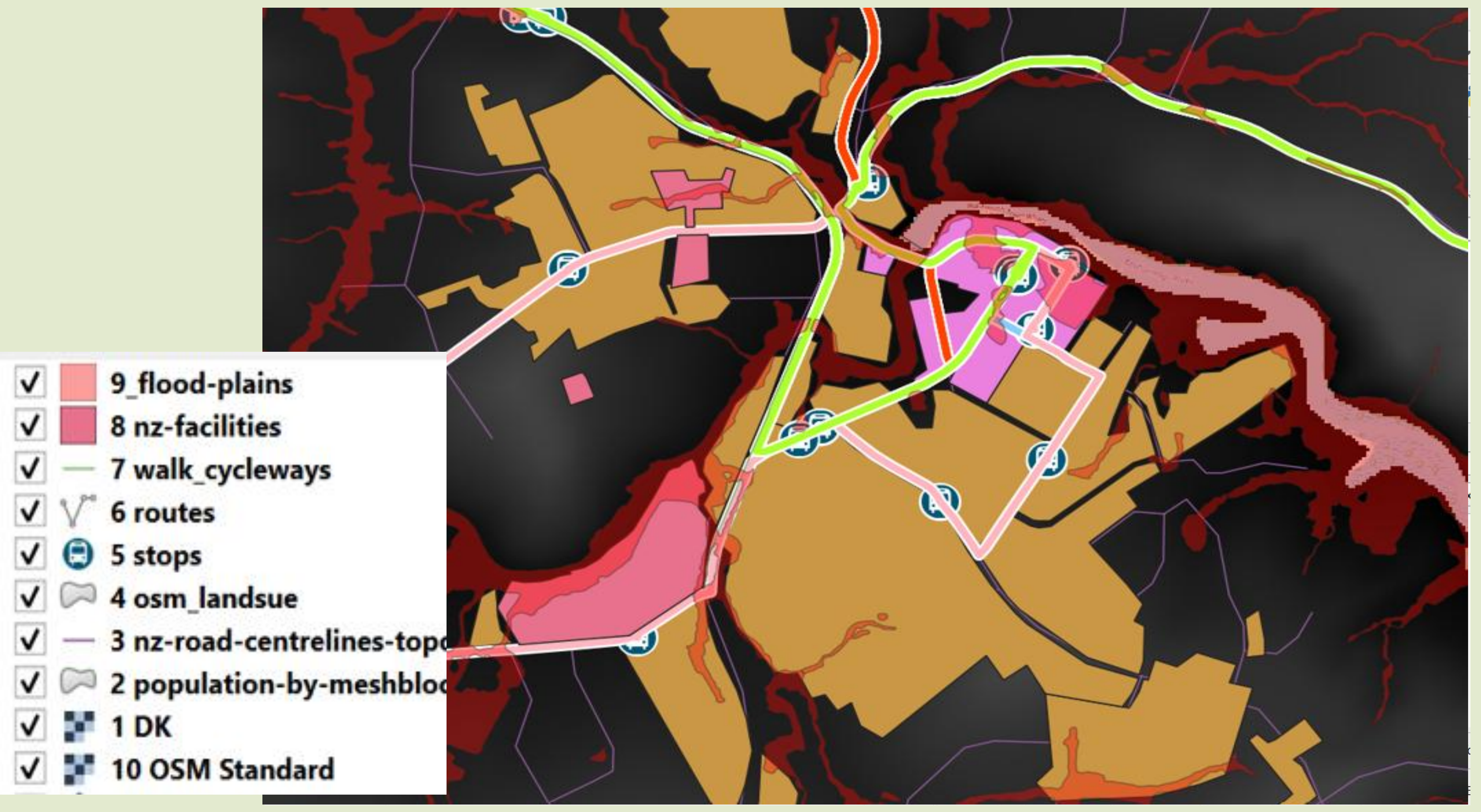
Suitable Site for a New Bus Station in Warkworth, NZ: A GIS-Based Multi-Criteria Analysis

Introduction

- ❑ Warkworth is growing rapidly as the northern expressway expands.
- ❑ With this growth, there's a pressing need to site a sustainable, accessible, and future-proof bus station.
- ❑ This project uses spatial analysis to identify the most suitable location.



Map 1: All dataset loaded (warkworth area)

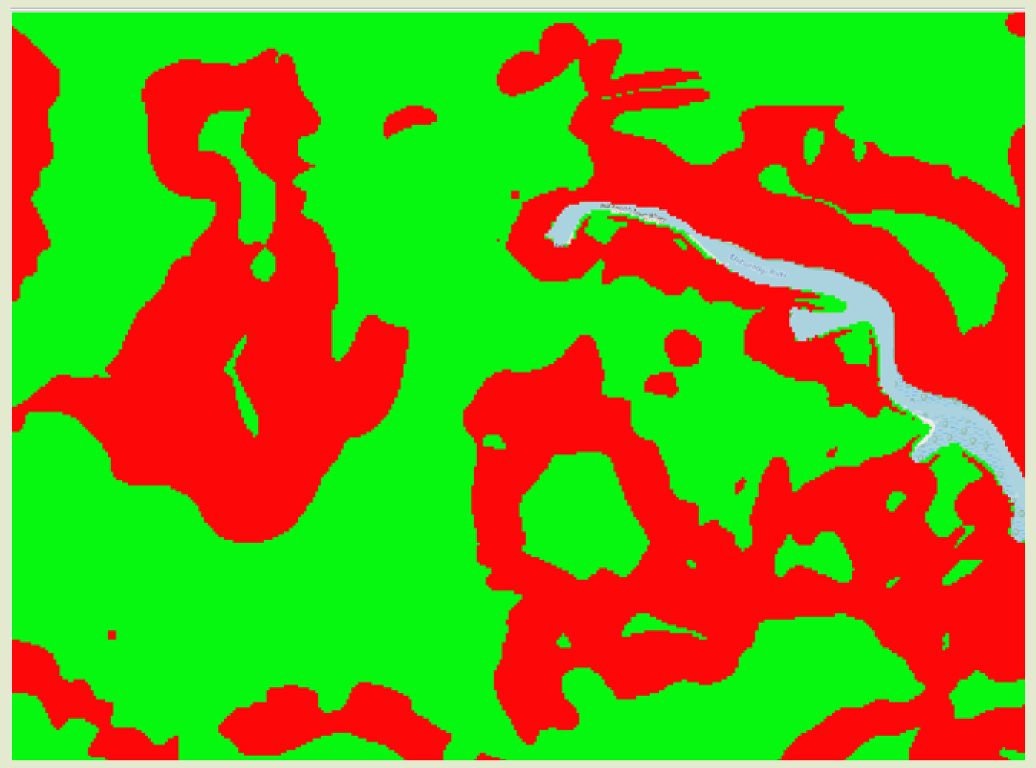
Datasets Used:

- Digital Elevation Model (8m resolution, LINZ)
- NZ Road Centrelines
- GTFS (bus routes and stops)
- OpenStreetMap Land Use and Basemap(web)
- Walk and Cycleways (Auckland Council)
- NZ Meshblock Population Data (Stats NZ)
- NZ Facilities (schools, hospitals)
- Floodplain Data (Auckland Council)

Workflow plan:

Need to spatially quantify factors like accessibility, safety, environmental suitability, and population proximity using a combination of raster and vector geoprocessing tools.

Methodology

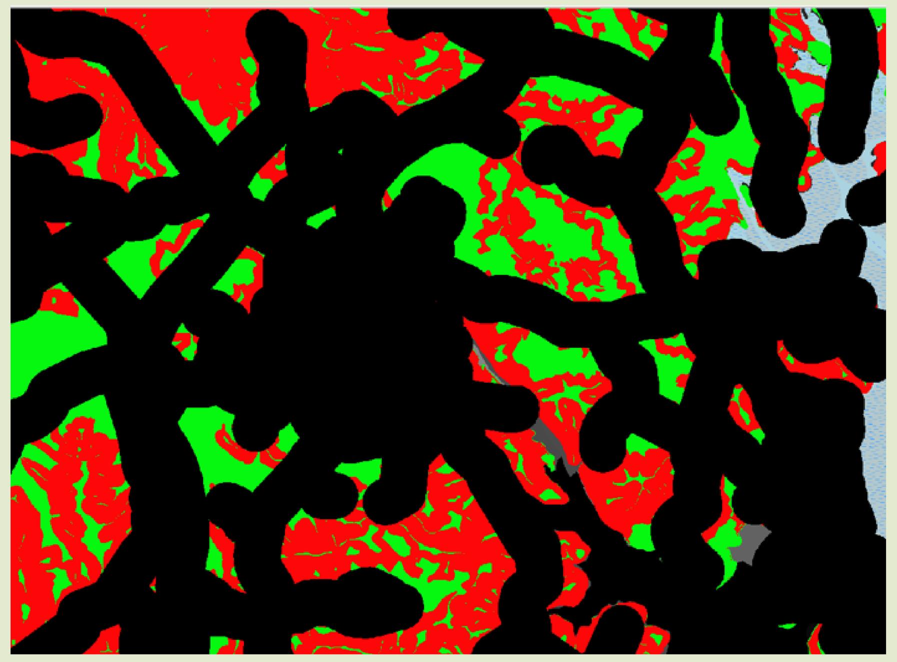


Map 2: Slope Analysis

Areas under 5° slope = Green (suitable)

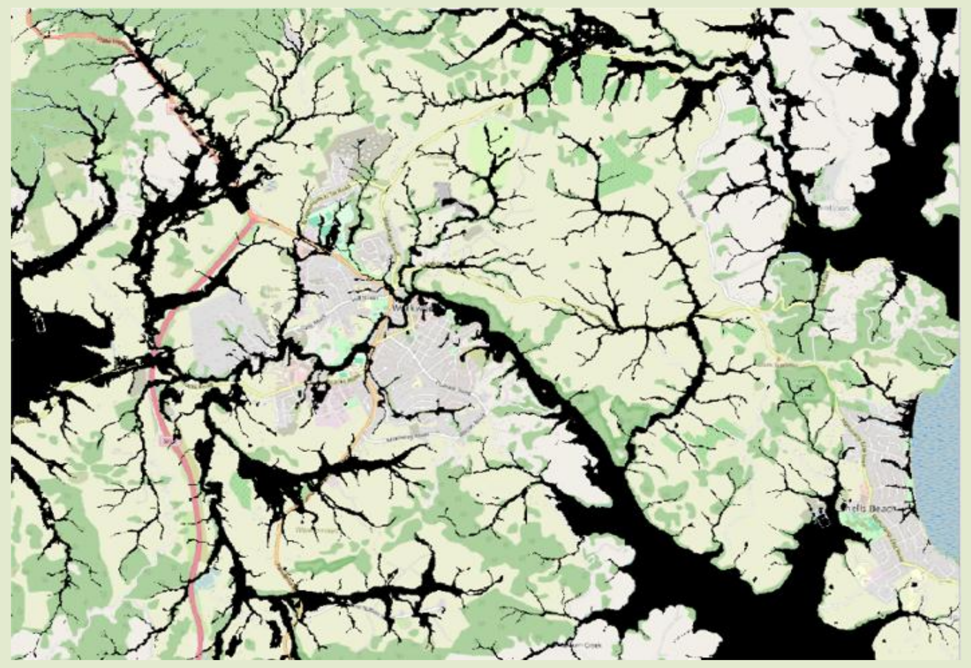
Map 3: Proximity Buffers

- ➢NZ Road Centrelines
- ➢GTFS Bus Stops
- ➢NZ Facilities
- ➢Walk / Cycleways



Map 4: Rasterized Layers

- ➢NZ Facilities
- ➢Road Centrelines
- ➢Bus Stops
- ➢Walk / Cycleways
- ➢Land use
- ➢Slope
- ➢Population meshblock



Map 5: Rasterized Flood Plains

Flood prone areas to exclude

Results

Map 6: Final Recommendations

Blue => High Suitable; Sky Blue => Less Suitable

Raster Calculation: Added all the factors with assigned justified weights and excluded flood prone areas

Raster Calculator Expression

```
( "roads_fin@1" *1.5  +  "population_fin@1" *2 + "landuse_fin@1" *1.5 +  "facilities_fin@1" *1.5 + "stops_fin@1" *1.5 + "slope_under5@1" *1 + "walkways_fin@1" *1) - ("floods_fin@1" *1.5)
```

Output - Most suitable zones:

- ✓ Near Falls Road, close to the Mahurangi River 📍 -36.402634, 174.658060
- ✓ A small pocket at west of Woodcocks Road 📍 -36.406774, 174.657379

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